

Fundamentals of Electrical Engineering: Series RLC Circuits & Transient Response Analysis

1. High-Level Overview & Basic Concepts

1.1 What is a Series RLC Circuit?

A **Series RLC Circuit** consists of three basic electrical components connected in series with a voltage source (V_S):

- **Resistor (R):** Dissipates energy in the form of heat (measured in Ohms, Ω).
- **Inductor (L):** Stores energy in a magnetic field (measured in Henries, H).
- **Capacitor (C):** Stores energy in an electric field (measured in Farads, F).

Because all elements are in series, the same instantaneous current $i(t)$ flows through every element, while the total supply voltage V_S splits across the components as $v_R(t)$, $v_L(t)$, and $v_C(t)$.

1.2 Physical Analogy

Think of a Series RLC circuit as a physical **mass-spring-damper system** (like a car suspension):

- **Capacitor (C):** Behaves like a **spring** (stores mechanical potential energy).
 - **Inductor (L):** Behaves like a **mass/inertia** (resists changes in motion/velocity).
 - **Resistor (R):** Behaves like a **shock absorber/friction** (dissipates energy to stop oscillations).
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1.3 Key Terminology

- **Transient Response:** The temporary, time-dependent behavior of the circuit immediately after a switch is closed or opened, before it settles into a steady state.
- **Natural Frequency (ω_0):** The rate at which the system naturally wants to oscillate when energy is released, given by:

$$\omega_0 = \frac{1}{\sqrt{LC}} \quad (\text{rad/s})$$

- **Damping Rate (α):** The rate at which energy is dissipated by the resistor, given by:

$$\alpha = \frac{R}{2L} \quad (1/\text{s})$$

- **Damped Frequency (ω_d):** The actual frequency at which the circuit oscillates in an underdamped condition:

$$\omega_d = \sqrt{\omega_0^2 - \alpha^2}$$

- **BIBO Stability: Bounded Input, Bounded Output.** A system is stable if any finite input produces a finite (non-infinite) output.
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2. Derivation of the Governing Differential Equation

2.1 Circuit Diagram & KVL Setup

 Diagram: Series RLC Circuit

Description of Visual Layout: A single-loop schematic featuring a DC voltage source V_S connected in series with a knife switch, a resistor R , an inductor L , and a capacitor C . Polarities for component voltages v_R , v_L , and v_C are marked, and current i flows clockwise through the loop.

2.2 Mathematical Derivation

Applying Kirchhoff's Voltage Law (KVL) around the loop after closing the switch at $t = 0$:

$$V_S = v_R(t) + v_L(t) + v_C(t) \quad \text{--- (1)}$$

Using the fundamental terminal relationships of electrical components:

1. Resistor voltage: $v_R(t) = i(t)R$
2. Inductor voltage: $v_L(t) = L \frac{di(t)}{dt}$
3. Capacitor current relation: $i(t) = C \frac{dv_C(t)}{dt}$ --- (2)

Substitute equation (2) into the voltage expressions for v_R and v_L :

- $v_R(t) = RC \frac{dv_C}{dt}$
- $v_L(t) = L \frac{d}{dt} \left(C \frac{dv_C}{dt} \right) = LC \frac{d^2 v_C}{dt^2}$

Substitute these into equation (1):

$$V_S = RC \frac{dv_C}{dt} + LC \frac{d^2 v_C}{dt^2} + v_C$$

Divide the entire equation by LC to obtain the standard second-order linear differential equation:

$$\frac{d^2 v_C}{dt^2} + \frac{R}{L} \frac{dv_C}{dt} + \frac{1}{LC} v_C = \frac{1}{LC} V_S \quad \text{--- (3)}$$

3. General Solution Structure

The complete solution to a non-homogeneous linear second-order differential equation consists of two parts:

$$v_C(t) = v_{c.f.}(t) + v_{p.i.}$$

- **Complementary Function ($v_{c.f.}$):** Represents the natural transient response of the system as it decays over time.
- **Particular Integral ($v_{p.i.}$):** Represents the forced response (steady-state value). For a constant DC input V_S , the particular integral is:

$$v_{p.i.} = V_S$$

3.1 Characteristic Equation

To find $v_{c.f.}(t)$, assume a solution of the form $v_C(t) = Ae^{st}$. Substituting this trial solution yields the **characteristic equation**:

$$s^2 + \frac{R}{L}s + \frac{1}{LC} = 0$$

Rewriting using damping rate $\alpha = \frac{R}{2L}$ and natural frequency $\omega_0 = \frac{1}{\sqrt{LC}}$:

$$s^2 + 2\alpha s + \omega_0^2 = 0$$

Using the quadratic formula, the roots s_1 and s_2 are:

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$$

4. Analysis of Response Types (Decided by Resistance R)

The physical behavior of the circuit depends entirely on the discriminant $(\alpha^2 - \omega_0^2)$, which is governed by the value of resistance R .

Response Type	Condition	Roots s_1, s_2	Voltage Response Equation $v_C(t)$	System Stability
Critically Damped	$\alpha = \omega_0$	Real and equal ($s_1 = s_2 = -\alpha$)	$v_C(t) = e^{-\alpha t}(A_1 t + A_2) + V_S$	Stable (Fastest settling without overshoot)
Overdamped	$\alpha > \omega_0$	Real and distinct	$v_C(t) = (A_1 e^{-\alpha_1 t} + A_2 e^{-\alpha_2 t}) + V_S$	Stable (Sluggish response, no oscillation)
Underdamped	$\alpha < \omega_0$	Complex conjugate ($s_{1,2} = -\alpha \pm j\omega_d$)	$v_C(t) = e^{-\alpha t}(A_1 \cos(\omega_d t) + A_2 \sin(\omega_d t)) + V_S$	Stable (Oscillates while decaying to steady state)
Undamped (Oscillator)	$R = 0 \implies \alpha = 0$	Purely imaginary ($s_{1,2} = \pm j\omega_0$)	$v_C(t) = A_1 \cos(\omega_0 t) + A_2 \sin(\omega_0 t) + V_S$	Marginally Stable (Sustained endless oscillations)
Unstable	$R < 0 \implies \alpha < 0$	Real parts positive	$v_C(t) \propto e^{+\alpha t} \dots$	Unstable (Uncontrolled exponential growth)

4.1 Detailed Breakdown of Each Case

Case 1: Critically Damped Response ($\alpha = \omega_0$)

- Resistance Condition:** $R = 2\sqrt{\frac{L}{C}}$
- Behavior:** The system returns to steady state as quickly as possible without oscillating or overshooting. Ideal for control systems where overshoot must be avoided.

Case 2: Overdamped Response ($\alpha > \omega_0$)

- Resistance Condition:** $R > 2\sqrt{\frac{L}{C}}$
- Behavior:** The energy is dissipated quickly by the high resistance. The voltage slowly approaches V_S without any oscillations.

Case 3: Underdamped Response ($\alpha < \omega_0$)

- Resistance Condition:** $R < 2\sqrt{\frac{L}{C}}$
- Behavior:** The stored energy bounces back and forth between the inductor and capacitor before being fully dissipated by R . This results in decaying sinusoidal oscillations at frequency $\omega_d = \sqrt{\omega_0^2 - \alpha^2}$.

4.2 Special Cases: Pure Oscillation & Instability

Case 4: Undamped Oscillator ($R = 0$)

Diagram: Undamped LC Oscillator Circuit and Waveforms

Description of Visual Layout: Left: Circuit diagram showing a DC source V_S , switch, Inductor L , and Capacitor C without any resistor ($R = 0$). Right: Time-domain plots showing constant-amplitude sinusoidal oscillations for capacitor voltage $v_C(t)$, inductor voltage $v_L(t)$, and capacitor current $i(t)$.

- Characteristic Equation:** $s^2 + \frac{1}{LC} = 0 \implies s = \pm j\omega_0$
- Physical Significance:** Energy continually sloshes back and forth between the magnetic field of L and the electric field of C with zero thermal loss. The circuit behaves as a pure oscillator producing continuous sine waves.

Case 5: Negative Resistance / Unstable System ($R < 0$)

 Diagram: Negative Resistance Waveform Growth

Description of Visual Layout: Graph displaying capacitor voltage versus time. After $t = 0.2$ ms, the waveform exhibits sinusoidal oscillations whose peak-to-peak amplitude exponentially grows larger over time rather than decaying.

- **Characteristic Equation:** $s^2 + \left(-\frac{R}{L}\right)s + \frac{1}{LC} = 0$
 - **Physical Significance:** A negative real part of α translates mathematically to an expanding exponential factor ($e^{+\alpha t}$). The system continuously injects energy into the circuit, causing uncontrolled growing oscillations that lead to instability.
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4.3 Summary Comparison of Response Waveforms

 Diagram: 2nd Order System Transient Responses Comparison

Description of Visual Layout: Comparative transient response curves over time (0 to 1.5 ms):

1. **Underdamped** ($R = 500 \Omega$): Overshoots the final level, oscillates, and eventually settles.
 2. **Critically Damped** ($R = 2000 \Omega$): Reaches steady state quickly without overshoot.
 3. **Overdamped** ($R = 4000 \Omega$): Slowly rises to steady state without overshoot.
 4. **Undamped** ($R = 0 \Omega$): Oscillates continuously with constant peak amplitude.
 5. **Unstable** ($R < 0$): Oscillates with exponentially increasing amplitude.
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5. Step-by-Step Worked Examples

Problem 1: Initial/Final State & Critical Resistance Calculation

Problem Statement:

An RLC series circuit is initially relaxed (all initial voltages and currents are zero). At $t = 0$ s, a switch connects a DC voltage source $V_S = 5$ V. The circuit parameters are $L = 1$ mH (1×10^{-3} H) and $C = 1$ μ F (1×10^{-6} F).

 Diagram: Illustration Problem Circuit Schema

Description of Visual Layout: Series RLC circuit containing a 5 V DC supply, switch closed at $t = 0$, variable resistor R , 1 mH inductor, and 1 μ F capacitor.

Evaluate:

1. $v_C(0^+)$ and $v_C(\infty)$
 2. $i_L(0^+)$ and $i_L(\infty)$
 3. The value of R required for a **critically damped response**.
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Solution:

Part (i): Finding Capacitor Voltages $v_C(0^+)$ and $v_C(\infty)$

1. **At $t = 0^+$ (Immediately after switch closure):**
 - Fundamental property of a capacitor: Voltage across a capacitor **cannot change instantaneously** because $i_C = C \frac{dv_C}{dt}$.
 - Since the circuit was initially relaxed ($v_C(0^-) = 0$ V):

$$v_C(0^+) = v_C(0^-) = 0 \text{ V}$$

2. **At $t = \infty$ (DC Steady State):**

- In a DC steady-state condition, the capacitor acts as an **open circuit** ($\frac{dv_C}{dt} = 0 \implies i = 0$).

- Zero current means zero voltage drop across R and L .
- Therefore, the entire supply voltage appears across the capacitor:

$$v_C(\infty) = V_S = 5 \text{ V}$$

Part (ii): Finding Inductor Currents $i_L(0^+)$ and $i_L(\infty)$

1. At $t = 0^+$ (Immediately after switch closure):

- Fundamental property of an inductor: Current through an inductor **cannot change instantaneously** because $v_L = L \frac{di}{dt}$.
- Since the circuit was relaxed ($i_L(0^-) = 0 \text{ A}$):

$$i_L(0^+) = i_L(0^-) = 0 \text{ A}$$

2. At $t = \infty$ (DC Steady State):

- As established in Part (i), the fully charged capacitor acts as an open circuit, blocking current flow through the series path.
- Therefore:

$$i_L(\infty) = 0 \text{ A}$$

Part (iii): Resistance R for Critical Damping

For a critically damped system, the damping rate α must equal the natural frequency ω_0 :

$$\alpha = \omega_0$$

Substitute definitions:

$$\frac{R}{2L} = \frac{1}{\sqrt{LC}}$$

Solve for R :

$$R = 2\sqrt{\frac{L}{C}}$$

Substitute the given numerical values ($L = 1 \times 10^{-3} \text{ H}$, $C = 1 \times 10^{-6} \text{ F}$):

$$R = 2\sqrt{\frac{1 \times 10^{-3}}{1 \times 10^{-6}}} = 2\sqrt{1000} = 2 \times 31.6227 = 63.25 \Omega$$

Final Calculated Values:

- $v_C(0^+) = 0 \text{ V}$
- $v_C(\infty) = 5 \text{ V}$
- $i_L(0^+) = 0 \text{ A}$
- $i_L(\infty) = 0 \text{ A}$
- $R_{\text{critical}} = 63.25 \Omega$

6. Key Takeaways & Exam Summary Checklist

- **Second-Order System:** Transient behavior in a series RLC circuit is modeled by a 2nd order differential equation:

$$\frac{d^2 v_C}{dt^2} + 2\alpha \frac{dv_C}{dt} + \omega_0^2 v_C = \omega_0^2 V_S$$

- **Role of Resistance (R):** R determines energy dissipation and governs system classification:
 - $R > 2\sqrt{L/C} \implies$ Overdamped (Slow, non-oscillatory)
 - $R = 2\sqrt{L/C} \implies$ Critically Damped (Fastest non-oscillatory convergence)

- $R < 2\sqrt{L/C} \implies$ Underdamped (Decaying oscillation)
- $R = 0 \implies$ Undamped Oscillator (Continuous oscillation at ω_0)
- $R < 0 \implies$ Unstable System (Exponentially growing amplitude)
- **Continuity Rules for Initial Conditions:**
 - Capacitor Voltage cannot jump instantaneously: $v_C(0^+) = v_C(0^-)$
 - Inductor Current cannot jump instantaneously: $i_L(0^+) = i_L(0^-)$